# Anthropic Models on Amazon Bedrock

Module 1:

Generative AI and Claude Models on AWS

# Course Contents

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Module 5: Security, Governance, and Guardrails

Module 6: Practical Architecture and Deployment Patterns

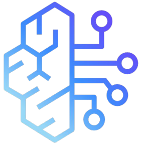
Module 7: Claude Code: Building Bedrock Applications

Module 8: Amazon Kiro and Enterprise Code Generation

# Module Objectives

## Claude Model Family

Identify Opus, Sonnet, and Haiku use cases based on intelligence, speed, and cost



## Amazon Bedrock Console

Access Claude models and enable model access in your AWS account

## Cost Optimization

Balance region selection, latency, model choice, and quotas for cost efficiency

## Value Propositions

Explain enterprise benefits: managed infrastructure, security, AWS integration

# What Makes Generative AI Different?

- Traditional ML: Classification from fixed categories
- Generative AI: Creates new content from patterns

“The key shift: from ‘Which box?’ to ‘What should I create?’”

# Why LLMs Matter for Enterprise

## Unstructured Data

Documents, emails, tickets, transcripts

## Translation

Business requests to technical specifications

## Natural Language

Understanding meaning at enterprise scale

## Workflow Power

Augment processes that needed humans

# Enterprise AI Requirements

1. **Control** — Content filtering, topic restrictions
2. **Integration** — Proprietary data, APIs, systems
3. **Security** — Encryption, IAM, audit trails
4. **Operations** — Cost tracking, latency, quotas

# The Claude Model Family

# The Claude Model Family

| Opus 4.6 | Sonnet 4.6 | Haiku 4.5 |
|---|---|---|
| Most intelligent<br>For complex tasks<br>$5/$25 per MTok | Best balance of<br>speed and intelligence<br>$3/$15 per MTok | Fastest model for<br>high-volume tasks<br>$1/$5 per MTok |

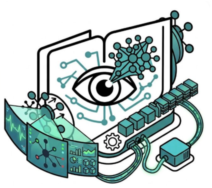
# Claude Model Specifications

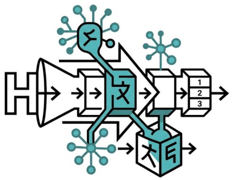
| Opus 4.6 | Sonnet 4.6 | Haiku 4.5 |
|---|---|---|
| $5/$25 per MTok | $3/$15 per MTok | $1/$5 per MTok |
| 1M token context | 1M token context | 200k token context |
| 128k max output | 64k max output | 64k max output |
| Highest intelligence | Balanced default | Fastest, cheapest |

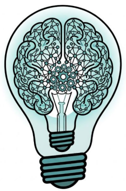
# When to Use Each Model

| Opus | Sonnet | Haiku |
|---|---|---|
| Legal analysis | Customer chatbots | Classification |
| Research synthesis | RAG applications | Moderation |
| Agent planning | Most workloads | Front-door routing |
| Complex reasoning | General purpose | High-volume tasks |

# Shared Claude Capabilities







| Vision | Streaming | Multilingual | Tools and Thinking |
| --- | --- | --- | --- |
| Photos, screenshots, diagrams as input | Real-time text output for UIs | Strong across many languages | Function calling, chain-of-thought |

# Model Selection Framework

| Requirement | Recommended Model |
|---|---|
| Maximum intelligence, complex reasoning | Opus 4.6 |
| Production workloads, balanced performance | Sonnet 4.6 |
| High volume, cost-sensitive, speed-critical | Haiku 4.5 |

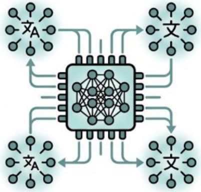
# Amazon Bedrock Overview

# Amazon Bedrock

Fully Managed Foundation Model Service

- Access Claude, Llama, Titan, and Stability AI from a single API
- Enterprise security with IAM, VPC endpoints, and CloudTrail
- No infrastructure to provision or manage—fully serverless
- Pay-per-token pricing with no upfront commitments

# Amazon Bedrock Value Proposition

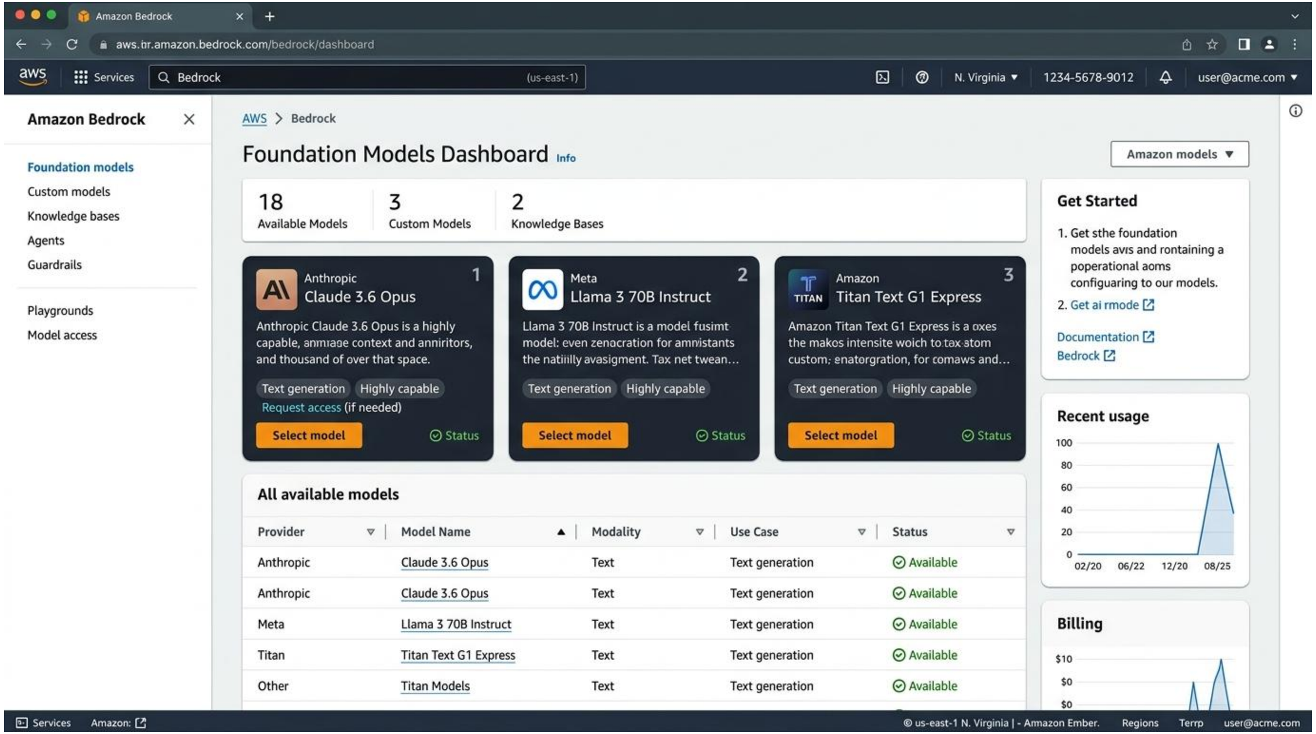



| Fully Managed | Multi-Model | Enterprise-Ready | AWS Integration |
|---|---|---|---|
| No infrastructure to maintain | Claude, Llama, Titan from one API | IAM, VPC, encryption built in | S3, Lambda, CloudWatch native |

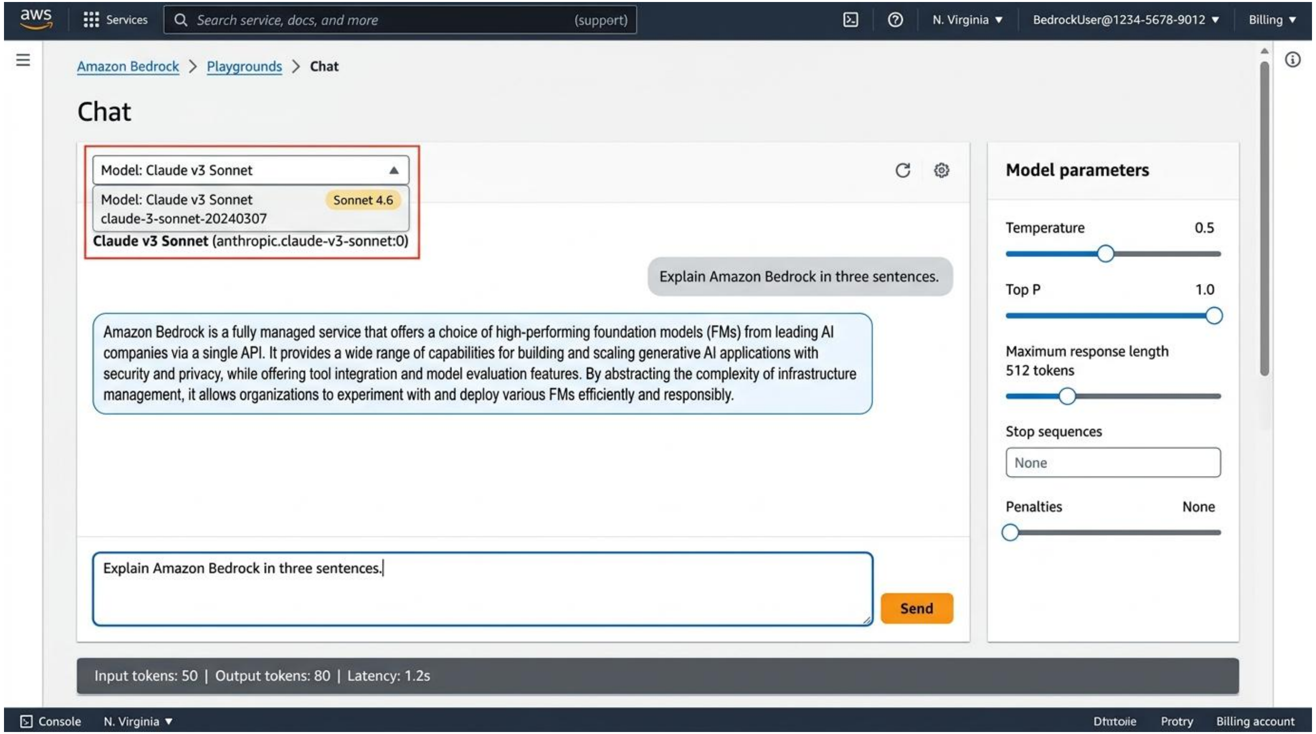
# Enterprise Security Features

1. **IAM integration:** Fine-grained access control
2. **VPC endpoints:** Keep traffic on your private network
3. **Encryption:** Data encrypted at rest and in transit
4. **CloudTrail:** Full audit logging of all API calls
5. **Data isolation:** Your data is never used to train models

# The Amazon Bedrock Console



# Demo: Amazon Bedrock Playground



# Claude Model Regional Availability

## AWS Americas

us-east-1, us-east-2 (Ohio)

us-west-1, us-west-2 (Oregon)

ca-central-1, sa-east-1

## AWS Europe and Asia

eu-west-1, eu-west-2, eu-central-1

ap-northeast-1 (Tokyo)

ap-southeast-1/2, me-south-1

# AWS GovCloud and Anthropic

Compliant Cloud Regions for Regulated Workloads

Claude Sonnet 4.5 runs in us-gov-west-1 and us-gov-east-1, supporting FedRAMP High compliance.

Isolated infrastructure for government workloads.

Other Claude models are coming soon to GovCloud.

# Region Selection Trade-offs

## Latency

Closer regions deliver faster responses; cross-ocean adds 80-100ms round-trip.

## Compliance

Data residency rules may force a specific region for regulated workloads.

## Pricing

Token pricing is consistent, but data transfer costs vary by region.

## Availability

Not all Claude models are in every region — verify before you commit.

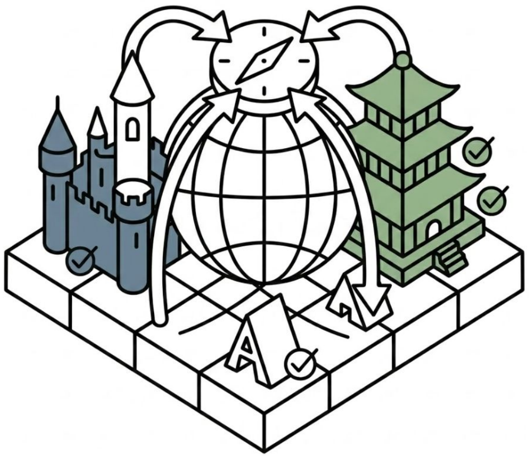
# Claude Model Pricing on Amazon Bedrock

| Model | Input (per MTok) | Output (per MTok) | Best For |
|---|---|---|---|
| Claude Opus 4.6 | $5 | $25 | Complex reasoning, agents |
| Claude Sonnet 4.6 | $3 | $15 | Balanced speed/intelligence |
| Claude Haiku 4.5 | $1 | $5 | High-volume, fast tasks |

# Cross-Region Inference Profiles

## Global Routing

- Global: Dynamic routing for max availability
- Regional: Guaranteed routing through a specific region
- Cross-region profiles for global deployment
- Trade-off: control over data location vs. uptime
- Choose based on compliance and SLA needs

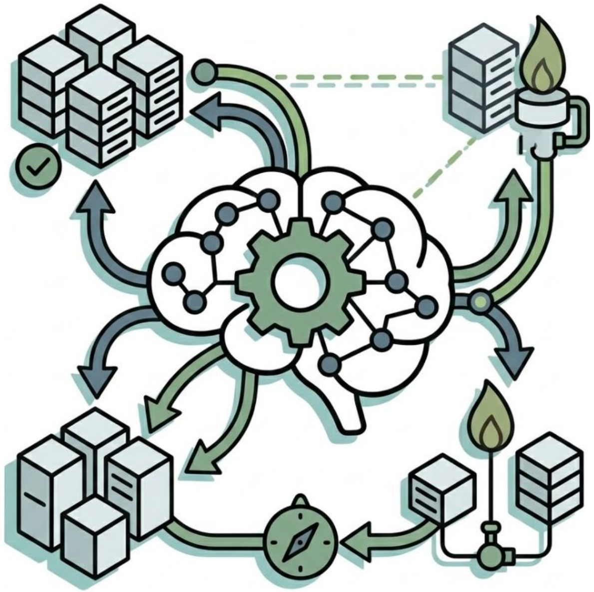


# Multi-Region Deployment Pattern

1. Primary region for normal operations (regional endpoint)
2. Secondary region for failover (regional endpoint)
3. Application-level health checks and routing
4. Cross-region inference for global availability tier

# Resilience Best Practices

- Implement exponential backoff with jitter for retries
- Monitor CloudWatch metrics for throttling and errors
- Test failover procedures regularly
- Consider warm standby vs. pilot light approaches



# What We Covered

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# Module 2 Is Next!

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